

NUCLEAR POWER AND THE THREE-STAGE PROGRAMME

FISSION-TO-GRID CHAIN → CRITICALITY MILESTONE LADDER → CORE-CONTROL FUNCTION MAP → THERMAL-FAST COMPARISON → FISSILE-FERTILE CONVERSION MAP → THREE-STAGE PROGRAMME RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

FISSION-TO-GRID CHAIN

FISSION-TO-GRID CHAIN

HEAVY NUCLEUS SPLITS AND RELEASES HEAT PLUS NEUTRONS

REMOVES CORE HEAT

STEAM SYSTEM

CONVERTS HEAT INTO WORKING-FLUID ENERGY

PRODUCES ELECTRICITY

REACTOR CAPACITY IS NOT ELECTRICITY GENERATED

STARTING CONDITION / INPUT

FISSION → heavy nucleus splits and releases heat plus neutrons

COOLANT → removes core heat

SCIENTIFIC OR ENGINEERING MECHANISM

STEAM SYSTEM → converts heat into working-fluid energy

TURBINE-GENERATOR → produces electricity

OUTPUT / FAILURE BOUNDARY

RULE → reactor capacity is not electricity generated

FISSION → heavy nucleus splits and releases heat plus neutrons

STARTING CONDITION / INPUT

• FISSION → heavy nucleus splits and releases heat plus neutrons

• COOLANT → removes core heat

SCIENTIFIC OR ENGINEERING MECHANISM

• STEAM SYSTEM → converts heat into working-fluid energy

• TURBINE-GENERATOR → produces electricity

OUTPUT / FAILURE BOUNDARY

• RULE → reactor capacity is not electricity generated

• FISSION → heavy nucleus splits and releases heat plus neutrons

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → reactor capacity is not electricity generated.

01

STAGE 01

CRITICALITY MILESTONE LADDER

CRITICALITY MILESTONE LADDER

CHAIN REACTION DECLINES

FIRST CRITICALITY

CONTROLLED SELF-SUSTAINING REACTION BEGINS

POWER ASCENSION

TESTED INCREASE TOWARD RATED CONDITIONS

GRID SYNCHRONISATION

GENERATOR CONNECTS TO GRID

SUBCRITICAL → chain reaction declines

FIRST CRITICALITY → controlled self-sustaining reaction begins

POWER ASCENSION → tested increase toward rated conditions

GRID SYNCHRONISATION → generator connects to grid

STARTING CONDITION / INPUT

• SUBCRITICAL → chain reaction declines

• FIRST CRITICALITY → controlled self-sustaining reaction begins

SCIENTIFIC OR ENGINEERING MECHANISM

• POWER ASCENSION → tested increase toward rated conditions

• GRID SYNCHRONISATION → generator connects to grid

OUTPUT / FAILURE BOUNDARY

• COMMERCIAL OPERATION → later declared operating status

• SUBCRITICAL → chain reaction declines

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POWER ASCENSION → tested increase toward rated conditions.

02

STAGE 02

CORE-CONTROL FUNCTION MAP

CORE-CONTROL FUNCTION MAP

CONTROL RODS

ABSORB NEUTRONS AND REGULATE REACTIVITY

SLOWS NEUTRONS IN A THERMAL REACTOR

TRANSFERS HEAT OUT OF THE CORE

BARRIER AGAINST RADIOACTIVE RELEASE

FOUR FUNCTIONS, FOUR DIFFERENT ANSWERS

SYSTEM / DEVICE / INSTITUTION

CONTROL RODS → absorb neutrons and regulate reactivity

MODERATOR → slows neutrons in a thermal reactor

DECISIVE TECHNICAL DISTINCTION

COOLANT → transfers heat out of the core

CONTAINMENT → barrier against radioactive release

STATUS / UNIT / EVIDENCE LIMIT

RULE → four functions, four different answers

CONTROL RODS → absorb neutrons and regulate reactivity

SYSTEM / DEVICE / INSTITUTION

• CONTROL RODS → absorb neutrons and regulate reactivity

• MODERATOR → slows neutrons in a thermal reactor

DECISIVE TECHNICAL DISTINCTION

• COOLANT → transfers heat out of the core

• CONTAINMENT → barrier against radioactive release

STATUS / UNIT / EVIDENCE LIMIT

• RULE → four functions, four different answers

• CONTROL RODS → absorb neutrons and regulate reactivity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOLANT → transfers heat out of the core.

03

STAGE 03

THERMAL-FAST COMPARISON

THERMAL-FAST COMPARISON

THERMAL REACTOR

MODERATED SLOWER-NEUTRON SPECTRUM

FAST REACTOR

NO MODERATOR AND FAST-NEUTRON SPECTRUM

HEAVY-WATER MODERATED STAGE-1 WORKHORSE

BREEDING OBJECTIVE WITH FERTILE BLANKET

COOLANT ALONE DOES NOT DEFINE THE SPECTRUM

SYSTEM / DEVICE / INSTITUTION

DECISIVE TECHNICAL DISTINCTION

STATUS / UNIT / EVIDENCE LIMIT

03

• CONTROL RODS → absorb neutrons and regulate reactivity

• MODERATOR → slows neutrons in a thermal reactor

• COOLANT → transfers heat out of the core

• CONTAINMENT → barrier against radioactive release

• RULE → four functions, four different answers

• CONTROL RODS → absorb neutrons and regulate reactivity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOLANT → transfers heat out of the core.

03

STAGE 03

THERMAL-FAST COMPARISON

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THERMAL REACTOR

MODERATED SLOWER-NEUTRON SPECTRUM

FAST REACTOR

NO MODERATOR AND FAST-NEUTRON SPECTRUM

HEAVY-WATER MODERATED STAGE-1 WORKHORSE

BREEDING OBJECTIVE WITH FERTILE BLANKET

COOLANT ALONE DOES NOT DEFINE THE SPECTRUM

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
THERMAL REACTOR → moderated slower-neutron spectrum	PHWR → heavy-water moderated Stage-1 workhorse	RULE → coolant alone does not define the spectrum
FAST REACTOR → no moderator and fast-neutron spectrum	FBR → breeding objective with fertile blanket	THERMAL REACTOR → moderated slower-neutron spectrum

SYSTEM / DEVICE / INSTITUTION

• THERMAL REACTOR → moderated slower-neutron spectrum

• FAST REACTOR → no moderator and fast-neutron spectrum

DECISIVE TECHNICAL DISTINCTION

• PHWR → heavy-water moderated Stage-1 workhorse

• FBR → breeding objective with fertile blanket

STATUS / UNIT / EVIDENCE LIMIT

• RULE → coolant alone does not define the spectrum

• THERMAL REACTOR → moderated slower-neutron spectrum

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → coolant alone does not define the spectrum.

04

STAGE 04

FISSILE-FERTILE CONVERSION MAP

FISSILE-FERTILE CONVERSION MAP

FISSILE MATERIALS

NEUTRON CAPTURE

PU-239 PATHWAY

NEUTRON CAPTURE AND DECAY

U-233 PATHWAY

FERTILE MATERIAL

NEEDS IRRADIATION AND CONVERSION

U-235 / PU-239 / U-233 → fissile materials

U-238 → neutron capture → PU-239 pathway

TH-232 → neutron capture and decay → U-233 pathway

FERTILE MATERIAL → needs irradiation and conversion

STARTING CONDITION / INPUT

• U-235 / PU-239 / U-233 → fissile materials

• U-238 → neutron capture → PU-239 pathway

SCIENTIFIC OR ENGINEERING MECHANISM

• TH-232 → neutron capture and decay → U-233 pathway

• FERTILE MATERIAL → needs irradiation and conversion

OUTPUT / FAILURE BOUNDARY

• RULE → resource abundance is not ready reactor fuel

• U-235 / PU-239 / U-233 → fissile materials

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → resource abundance is not ready reactor fuel.

05

STAGE 05

THREE-STAGE PROGRAMME RAIL

THREE-STAGE PROGRAMME RAIL

STAGE 1

PHWR, NATURAL URANIUM AND PLUTONIUM-BEARING SPENT FUEL

SEPARATES MATERIAL FOR THE BREEDER BRIDGE

STAGE 2

FAST BREEDER AND FISSILE-INVENTORY MULTIPLICATION

THORIUM IRRADIATION

PREPARES U-233 ROUTE

STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel

REPROCESSING → separates material for the breeder bridge

STAGE 2 → fast breeder and fissile-inventory multiplication

THORIUM IRRADIATION → prepares U-233 route

STAGE 3 → long-term thorium-U-233 utilisation horizon

CONCEPT / ARCHITECTURE

• STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel

• REPROCESSING → separates material for the breeder bridge

MECHANISM / APPLICATION

• STAGE 2 → fast breeder and fissile-inventory multiplication

• THORIUM IRRADIATION → prepares U-233 route

READINESS / STATUS LIMIT

• STAGE 3 → long-term thorium-U-233 utilisation horizon

• STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STAGE 2 → fast breeder and fissile-inventory multiplication.

06

STAGE 06

PFBR STATUS CARD

PFBR STATUS CARD

IGCAR FAST-REACTOR AND SODIUM EXPERTISE

BHAVINI PROJECT RESPONSIBILITY

500 MWE DESIGN VALUE IN FETCHED DAE TEXT

6 APRIL 2026

FIRST CRITICALITY ACHIEVED

NOT ESTABLISHED

GRID SYNCHRONISATION OR COMMERCIAL OPERATION

DESIGN → IGCAR fast-reactor and sodium expertise

BUILD / COMMISSION → BHAVINI project responsibility

RATING → 500 MWe design value in fetched DAE text

6 APRIL 2026 → first criticality achieved

STARTING CONDITION / INPUT

• DESIGN → IGCAR fast-reactor and sodium expertise

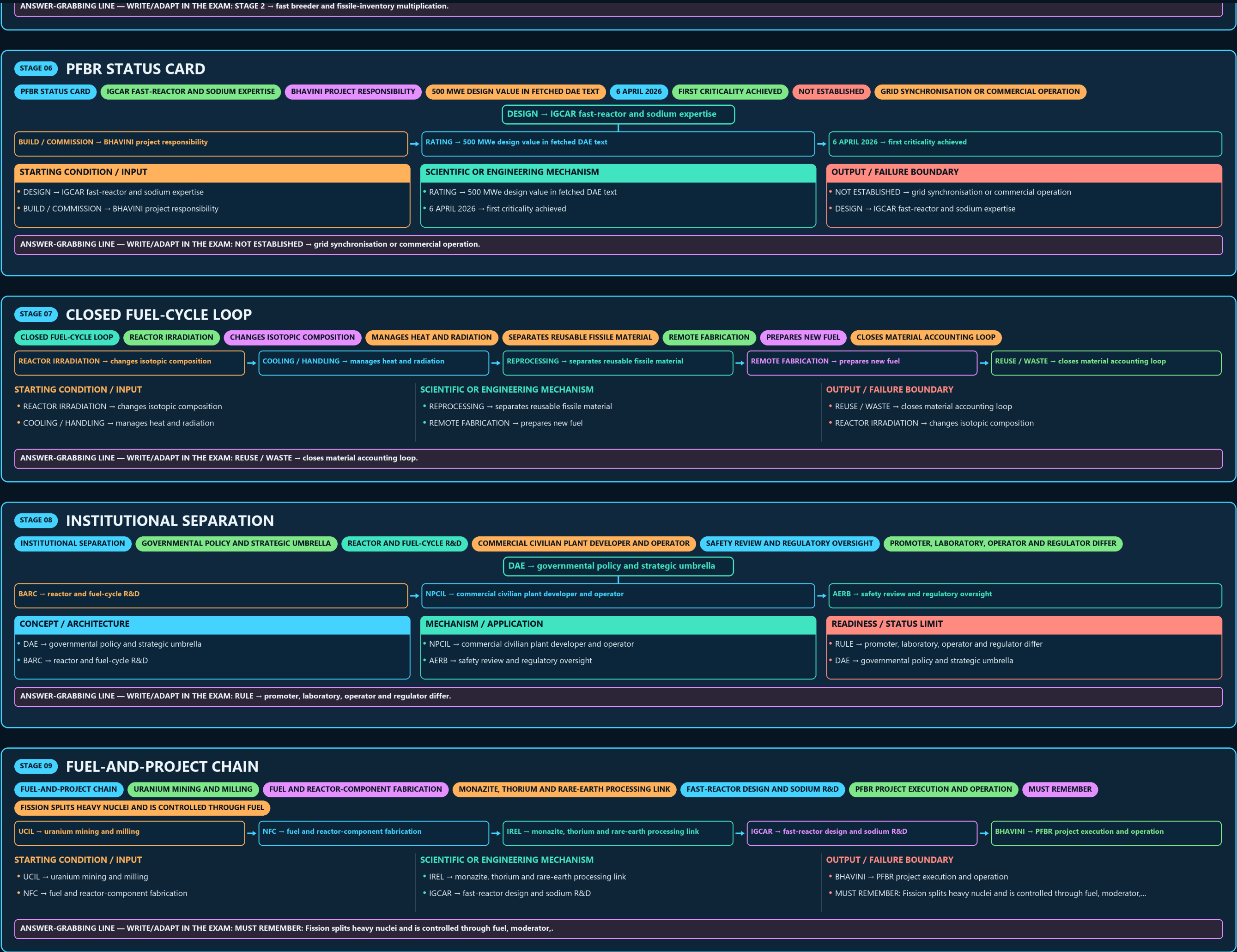
SCIENTIFIC OR ENGINEERING MECHANISM

• RATING → 500 MWe design value in fetched DAE text

OUTPUT / FAILURE BOUNDARY

• NOT ESTABLISHED → grid synchronisation or commercial operation

NUCLEAR POWER AND THE THREE-STAGE PROGRAMME • TILE 2/4 • same master rows 2564-5798 of 10926 • continues below



09

STAGE 09 FUEL-AND-PROJECT CHAIN

FUEL-AND-PROJECT CHAIN

URANIUM MINING AND MILLING

FUEL AND REACTOR-COMPONENT FABRICATION

MONAZITE, THORIUM AND RARE-EARTH PROCESSING LINK

FAST-REACTOR DESIGN AND SODIUM R&D

PFBR PROJECT EXECUTION AND OPERATION

MUST REMEMBER

FISSION SPLITS HEAVY NUCLEI AND IS CONTROLLED THROUGH FUEL

UCIL → uranium mining and milling

NFC → fuel and reactor-component fabrication

IREL → monazite, thorium and rare-earth processing link

IGCAR → fast-reactor design and sodium R&D

BHAVINI → PFBR project execution and operation

STARTING CONDITION / INPUT

- UCIL → uranium mining and milling
- NFC → fuel and reactor-component fabrication

SCIENTIFIC OR ENGINEERING MECHANISM

- IREL → monazite, thorium and rare-earth processing link
- IGCAR → fast-reactor design and sodium R&D

OUTPUT / FAILURE BOUNDARY

- BHAVINI → PFBR project execution and operation
- MUST REMEMBER: Fission splits heavy nuclei and is controlled through fuel, moderator,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Fission splits heavy nuclei and is controlled through fuel, moderator,.

10

STAGE 10 REACTOR TAXONOMY

REACTOR TAXONOMY

HEAVY WATER AND NATURAL-URANIUM STAGE-1 ROLE

LIGHT-WATER SYSTEMS WITH DIFFERENT STEAM CYCLES

FAST SPECTRUM AND BREEDING OBJECTIVE

SMALL-REACTOR LINES

DESIGN OR DEVELOPMENT STATUS

TYPE AND DEPLOYMENT STATUS MUST BE STATED TOGETHER

CLOSE DISTINCTION

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PHWR → heavy water and natural-uranium Stage-1 role	FBR → fast spectrum and breeding objective	RULE → type and deployment status must be stated together
PWR / BWR → light-water systems with different steam cycles	AHWR / SMALL-REACTOR LINES → design or development status	CLOSE DISTINCTION: Fissile is not fertile, enrichment is not reprocessing, heavy water is...

SYSTEM / DEVICE / INSTITUTION

- PHWR → heavy water and natural-uranium Stage-1 role
- PWR / BWR → light-water systems with different steam cycles

DECISIVE TECHNICAL DISTINCTION

- FBR → fast spectrum and breeding objective
- AHWR / SMALL-REACTOR LINES → design or development status

STATUS / UNIT / EVIDENCE LIMIT

- RULE → type and deployment status must be stated together
- CLOSE DISTINCTION: Fissile is not fertile, enrichment is not reprocessing, heavy water is...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → type and deployment status must be stated together.

11

STAGE 11 NUCLEAR ANSWER SPINE

NUCLEAR ANSWER SPINE

FISSION, CRITICALITY, FISSILE, FERTILE AND FUEL CYCLE

STAGE 1, PLUTONIUM BRIDGE, STAGE 2 AND THORIUM HORIZON

DAE BARC NPCIL AERB IGCAR BHAVINI AND FUEL CHAIN

CAPACITY CRITICALITY GRID GENERATION AND LEGAL STATUS

DATED FLEET FUEL REACTOR LAW COST AND SCHEDULE EVIDENCE

EVIDENCE LIMIT

TRACE PHWR URANIUM, FAST-BREEDER

DEFINE → fission, criticality, fissile, fertile and fuel cycle

TRACE → Stage 1, plutonium bridge, Stage 2 and thorium horizon

MAP → DAE BARC NPCIL AERB IGCAR BHAVINI and fuel chain

SEPARATE → capacity criticality grid generation and legal status

VERIFY → dated fleet fuel reactor law cost and schedule evidence

CONCEPT / ARCHITECTURE

- DEFINE → fission, criticality, fissile, fertile and fuel cycle
- TRACE → Stage 1, plutonium bridge, Stage 2 and thorium horizon

MECHANISM / APPLICATION

- MAP → DAE BARC NPCIL AERB IGCAR BHAVINI and fuel chain
- SEPARATE → capacity criticality grid generation and legal status

READINESS / STATUS LIMIT

- VERIFY → dated fleet fuel reactor law cost and schedule evidence
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Trace PHWR uranium, fast-breeder...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Trace PHWR uranium, fast-breeder...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Trace PHWR uranium, fast-breeder.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

POLICY, STRATEGIC DIRECTION AND OVERALL NUCLEAR-GOVERNANCE UMBRELLA.

R&D BACKBONE FOR REACTOR TECHNOLOGY, FUEL CYCLE AND ADVANCED

THE FAST-REACTOR AND SODIUM-TECHNOLOGY DESIGN CENTRE — IT DESIGNED

BHAVINI IS THE SEPARATE COMPANY THAT BUILT IT AND

CONFLATING IGCAR WITH BHAVINI IS A COMMON INSTITUTIONAL ERROR.

COMMERCIAL REACTOR OPERATOR/DEVELOPER FOR THE CIVILIAN FLEET

ALSO THE VEHICLE FOR THE BHARAT SMALL REACTOR (220

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- DAE: policy, strategic direction and overall nuclear-governance umbrella.
- BARC: R&D backbone for reactor technology, fuel cycle and advanced concepts (including the AHWR design and the BSMR line).
- IGCAR: the fast-reactor and sodium-technology design centre — it designed PFBR
- BHAVINI is the separate company that built it and will operate it.

READINESS, UNIT OR STATUS LIMIT

- Conflating IGCAR with BHAVINI is a common institutional error.
- NPCIL: commercial reactor operator/developer for the civilian fleet
- also the vehicle for the Bharat Small Reactor (220 MWe PHWR for captive industrial use) solicitation.
- BHAVINI: Stage 2 breeder-reactor implementation body linked to PFBR.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- NFC / UCIL / IREL: fuel fabrication, uranium mining, and monazite processing for
- thorium and rare earths — the physical supply chain beneath the three-stage narrative.
- AERB: technical-safety regulator, distinct from liability legislation — and, as of the verified position, not a statutorily independent authority .
- CAUTION: A major debate is how to balance ambitious capacity expansion with

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.